

# Fast Iterative Solvers. Project 1.

Summer Semester 2025

Douglas R.Q. Pacheco / Nicolás Espinoza Contreras

**Due: Monday 6th July, 2026, 23:59 hrs.**

*The project may be completed individually or in pairs. However, if it is developed in pairs, **each participant must submit their own copy** of the project to Moodle, including both names on the submission.*

## Overview

The main objective of this project is to identify and evaluate the variations among different Krylov subspace approaches in a given application and their behavior under specific circumstances. Its main objectives are understanding runtime, the orthogonality of the Krylov subspace, the number of iterations (i.e., Krylov vectors generated), and the effect of preconditioning on the problem. You are asked to implement the following Krylov subspace methods:

- GMRES with preconditioner,
- Conjugate Gradient (CG) method,

to solve a linear system  $Ax = b$ , where  $A$  is a real-valued square matrix (symmetric positive definite in the case of CG), and  $b$  is a given vector.

## General Instructions

Download the matrices

- `gmres_matrix_msr.txt`
- `cg_matrix_msr_1.txt` and `cg_matrix_msr_2.txt`

from the project page on Moodle. These matrices are stored in *modified compressed sparse row* (MSR) format. Some annotations to help read the files are provided along with the assignment.

For all tests, you should use the following setup:

- Prescribe the solution vector  $x = (1, 1, \dots, 1)^T$ , and determine the corresponding right-hand side  $b = Ax$ .
- Use the initial guess  $x_0 = 0$ .
- Use a tolerance of  $\|r_k\|_2 / \|r_0\|_2 = 10^{-8}$  to establish convergence. Whenever the relative residual drops below this value, we consider the iteration to be converged.
- **Important:** Based on the previous point, for preconditioned GMRES, the residual is also preconditioned, i.e.,  $r_k = M^{-1}(b - Ax_k)$ . For restarted GMRES, the iteration index  $k$  represents the cumulative iteration count, that is, the total number of Krylov vectors generated.

As we determine the right-hand-side  $b$  such that it corresponds to a known solution  $x$ , we can also compute the error  $e_k = x_k - x$  at each iteration  $k$ .

## Specific Instructions

### GMRES Method

- Consider the matrix `gmres_matrix_msr` for the current section.
- The GMRES algorithm should be implemented in restarted formulation GMRES( $m$ ). Full GMRES can be tested by choosing a restart parameter large enough.
- Apply **left** preconditioning to the GMRES procedure. Implement the following options:
  1. Jacobi preconditioning;
  2. Gauss–Seidel preconditioning;
  3. ILU(0) preconditioning.
- For the full GMRES method, with and without preconditioning, plot the relative residual against iteration index on a semi-log scale. *Compare and analyze the results obtained based on how many Krylov vectors were needed to converge in each case.*
- For full GMRES (without preconditioning): *check the orthogonality of the Krylov vectors.*  
**Hints:** Plot the computed values of  $\mathbf{v}_1 \cdot \mathbf{v}_k$  against the number of iterations on a semi-log scale. *Explain what you observe.*
- For GMRES( $m$ ), *Compare* the number of Krylov spaces and the runtime required for the following restart parameters  $m = 10, 25, 50, 200$  and with the full GMRES, and *explain what you can observe.*
- Set the restart number to  $m=200$ , and compare the cases applying the three preconditioners with the case without preconditioner (GMRES(200)). Compare the number of iterations needed to converge and runtime (computational resources needed would be ideal to show as well, but it is not mandatory.). *Comment about the results.*  
**Hints:** Does the runtime increase or decrease compared to full GMRES? If the runtime changes, why does this happen? What factors, other than runtime, might motivate the use of restarts instead of full GMRES?

### Conjugate Gradient Method

- Consider the matrices `cg_matrix_msr_1` and `cg_matrix_msr_2` for the current section.
- The CG method should be implemented as discussed in class. **You will not implement preconditioning for this case.**
- Plot the *error* in A-norm, i.e.,  $\|\mathbf{e}\|_A = \sqrt{\langle \mathbf{A}\mathbf{e}, \mathbf{e} \rangle}$ , as well as the *residual* and *relative residual* in standard 2-norm, i.e.,  $\|\mathbf{r}\|_2 = \sqrt{\langle \mathbf{r}, \mathbf{r} \rangle}$ , against iteration index on a semi-log scale for both cases. *Compare qualitatively the difference in convergence between these three measurements. Comment on what you can observe.*
- *Compare the differences in convergence between the two CG cases. Comment on what is happening and why.*

## Report

You should write a short report that addresses all the points raised in the previous section. The report should be written in any digital platform (i.e., LaTeX, Word, among others), and it should have a brief introduction, brief theoretical part, results, discussion and conclusion. **This report should not exceed four pages (excluding figures).**